

CANDIDATE'S NAME: _____

CTG: _____

YISHUN JUNIOR COLLEGE
2017 JC2 PRELIMINARY EXAMINATION

CHEMISTRY
HIGHER 2

9729/03

Paper 3 Free Response Questions

MONDAY 11 SEPTEMBER 2017
1400hrs – 1600hrs
(2 hours)

Candidates answer in separate paper.

Additional Materials: Writing Papers
Data Booklet



READ THESE INSTRUCTIONS FIRST

Write your name and CTG on all the work that you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **one** question.

The use of an approved calculator is expected, where appropriate.
A Data Booklet is provided.

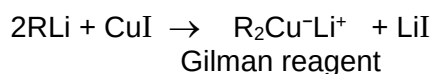
The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, fasten all your work for Section A **separately** from Section B with the respective cover sheet on top.

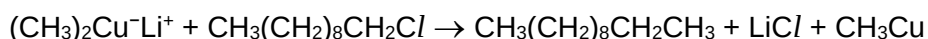
Section A

Answer **all** the questions in this section.

- 1 (a) Copper often occurs in the Earth's crust as copper(I) sulfide, Cu_2S , associated with the sulfides of zinc, ZnS , and silver, Ag_2S . The first stage of the extraction of copper from its ore involves the partial air-oxidation to copper(I) oxide, Cu_2O , and sulfur dioxide, SO_2 , followed by the reaction between this copper(I) oxide and unreacted copper(I) sulfide to give copper and more sulfur dioxide.
- (i) Write balanced chemical equations for these two reactions. [2]
- (ii) Suggest how the melting point of copper(I) sulfide might compare to that of copper(I) oxide. Explain your answer. [2]
- (iii) This first stage produces copper containing some zinc and silver as impurities. The solid impure copper is purified by electrolysis.
Describe, with reference to the E° values, the electrode reactions that take place during this electrolysis, and explain in detail how each of the two impurity metals is removed from the copper. [3]
- (iv) Using inert electrodes, a current was passed through two beakers containing aqueous silver nitrate and aqueous copper(II) sulfate, connected in series. After 30 minutes, 0.100 g of silver was deposited from the first solution.
Calculate the current passed and the mass of copper deposited from the aqueous copper(II) sulfate. [2]
- (b) When KI(aq) is added to a solution containing $\text{Cu}^{2+}(\text{aq})$, a white precipitate of the highly insoluble copper(I) iodide and a brown solution is formed.
- (i) By selecting appropriate E° values from the *Data Booklet*, explain why it would be expected that this redox reaction would **not** occur. [1]
- (ii) Suggest a possible reason for why it does in fact occur. [1]
- (c) Copper(I) iodide is used in the synthesis of Gilman reagents, $\text{R}_2\text{Cu}^-\text{Li}^+$, which are very useful in organic synthesis because larger molecules can be prepared from smaller ones.



For example, lithium dimethylcopper reacts with 1-chlorodecane to give undecane.

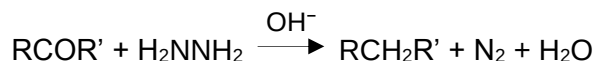


The above reaction is called organometallic coupling reaction.

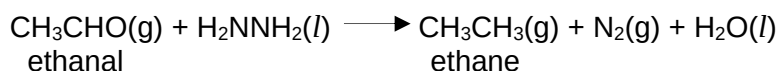
Similarly, organometallic coupling reaction can be used to synthesise ketones by reacting suitable Gilman reagents and acid chlorides.

Suggest a suitable Gilman reagent and an acid chloride that can be used to synthesise 5,6-dimethyldecan-3-one. [2]

- (d) Carbonyl compounds can be converted into alkanes by reacting the carbonyl compounds with hydrazine, H_2NNH_2 , in the presence of KOH . This reaction is named as the Wolff-Kishner reaction, which is named by the two chemists, Ludwig Wolff and N.M. Kishner, who discovered it in 1911.



For example, ethane can be synthesised from ethanal and hydrazine, according to the following equation.



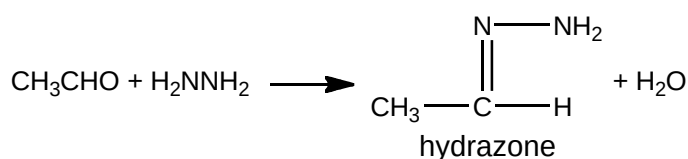
For this reaction, $\Delta S^\ominus = +191.5 \text{ J K}^{-1} \text{ mol}^{-1}$ at 298 K.

- (i) Explain what is meant by the term *entropy* of a chemical system. [1]
- (ii) Suggest what is the significance of the sign of the entropy change with respect to the reaction between ethanal and hydrazine. [1]

The following table lists some $\Delta H_f^\ominus$ values.

compound	$\Delta H_f^\ominus / \text{kJ mol}^{-1}$
$\text{CH}_3\text{CHO}(\text{g})$	-196.4
$\text{H}_2\text{NNH}_2(\text{l})$	+50.6
$\text{CH}_3\text{CH}_3(\text{g})$	-84.7
$\text{H}_2\text{O}(\text{l})$	-285.8

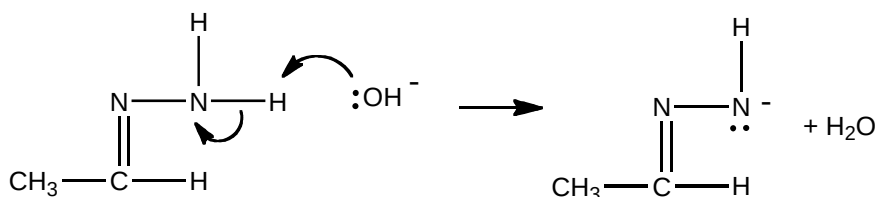
- (iii) Use the data above to calculate $\Delta H^\ominus$ for the reaction between ethanal and hydrazine. Hence calculate the $\Delta G^\ominus$ for the reaction between ethanal and hydrazine. Leave all your answers to one decimal place. [2]
- (iv) The first step of the Wolff-Kishner reaction involves the formation of a hydrazone intermediate.



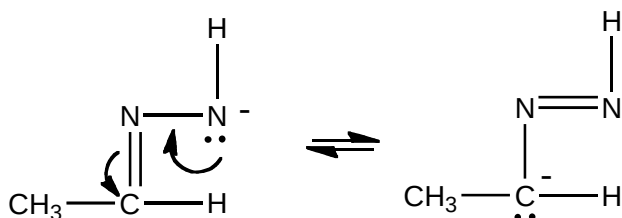
State the type of reaction that has occurred. [1]

- (v) After the formation of the hydrazone, the reaction occurs via a five-step mechanism to obtain ethane as the product.

step 1: abstraction of one of the weakly acidic protons from the -NH_2 group by OH^- , to form a hydrazone anion.



step 2: the hydrazone anion has a resonance form that places the negative charge on the carbon atom and has a double bond between the two nitrogen atoms.



step 3: protonation of the carbon atom of the resonance form of the hydrazone anion by H_2O to yield a neutral intermediate.

step 4: abstraction of a proton from the neutral intermediate by OH^- to form N_2 , an ethyl anion, CH_3CH_2^- , and H_2O .

step 5: protonation of the carbon atom of the ethyl anion by H_2O to yield ethane and OH^- .

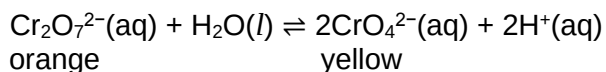
Using the information given above, suggest the mechanism for steps 3, 4 and 5.

Show any relevant lone pairs and charges, and indicate the movement of electron pairs with curly arrows. [3]

[Total: 21]

- 2 Manganate(VII) and dichromate(VI) ions are oxidising agents that are commonly used in organic synthesis.

(a) In aqueous solution, dichromate(VI) ions exist in equilibrium with chromate(VI) ions.



- (i) State Le Chatelier's Principle. [1]
- (ii) Explain why a solution of potassium chromate turns from yellow to orange on the addition of dilute hydrochloric acid. [1]

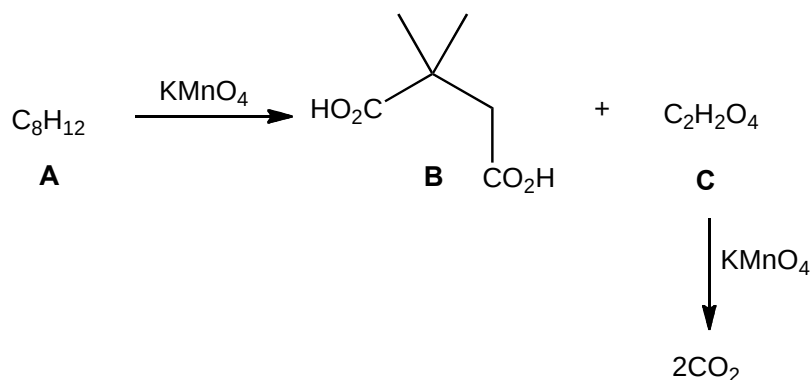
Following the addition of dilute hydrochloric acid, the solution was warmed. This caused the colour to change from orange back to yellow.

- (iii) State whether the forward reaction is exothermic or endothermic. [2]
Explain your answer.

At 25 °C, the equilibrium constant, K_c , for the above equilibrium has a value of 1.31×10^{-13} .

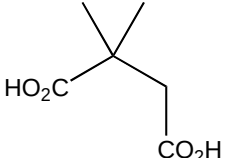
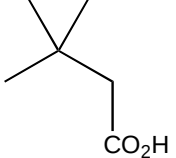
- (iv) Calculate a value for the ratio $\frac{[\text{CrO}_4^{2-}]^2}{[\text{Cr}_2\text{O}_7^{2-}]}$ at pH 8. [1]
- (v) Use your answer for (a)(iv) to deduce the colour of the solution at pH 8. [1]

(b) Hot, acidified concentrated potassium manganate(VII) can oxidise various functional groups to ketones, carboxylic acids and carbon dioxide, allowing the structures to be determined. The following scheme shows the reaction of compound **A**.

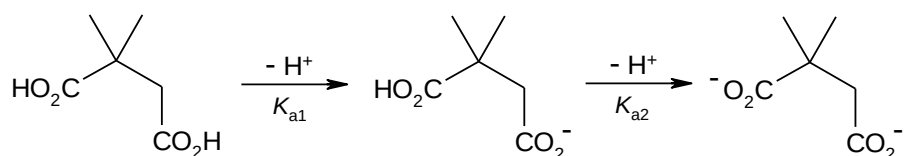


- (i) Suggest the structures of compounds **A** and **C**. [2]
- (ii) Explain why compound **A** does not exhibit *cis-trans* isomerism. [1]

(iii) Compound **B** and 3,3-dimethylbutanoic acid are both weak acids.

acid	formula
compound B	
3,3-dimethylbutanoic acid	

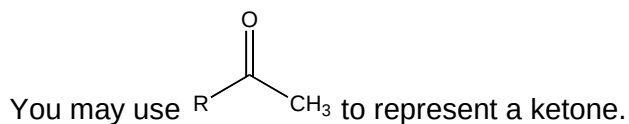
Compound **B** is a dicarboxylic acid which ionises in two stages.



Suggest a reason why the K_{a1} value of compound **B** is much higher than the K_a value of 3,3-dimethylbutanoic acid. [1]

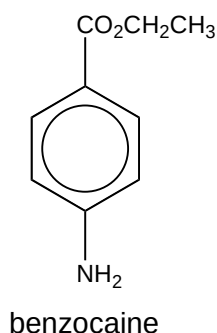
(c) Hydroxynitriles can be synthesised from ketones, and they are useful intermediates in the synthesis of amino acids.

Name and outline the mechanism for the synthesis of hydroxynitrile from a ketone. In your answer, show any relevant lone pairs, dipoles and charges, and use curly arrows to indicate the movement of electron pairs.



[3]

- (d) Benzocaine is well known for its local anaesthetic properties and is commonly used in antiseptic creams and sunburn remedies.



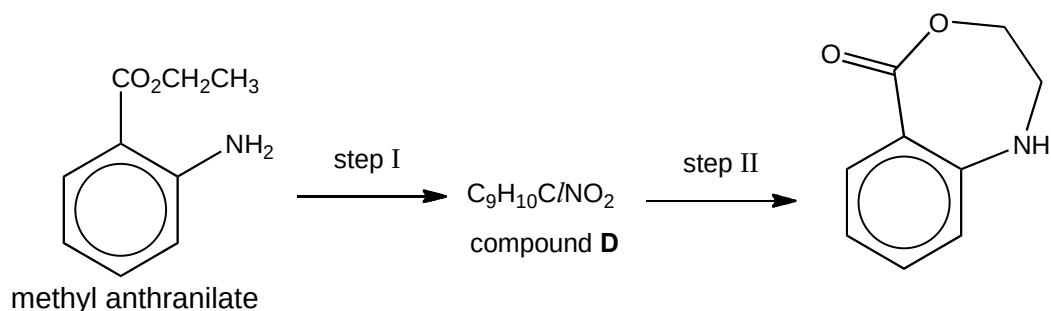
The preparation of benzocaine from methylbenzene involves the use of potassium manganate(VII) in one of the steps.

- (i) *Use of the Data Booklet is relevant to this question.*

Suggest how methylbenzene can be converted to benzocaine. Give reagents, conditions and the structural formulae of the compounds obtained for each stage of the conversion. [4]

- (ii) Describe a simple chemical test that could be used to distinguish methylbenzene from benzocaine, other than the use of KMnO_4 . State your observations clearly. [2]

- (e) Methyl anthranilate is an isomer of benzocaine and it is used as a bird repellent to protect crops. It can be used in the following organic synthesis.



- (i) Suggest the reagents and conditions needed for step I. [1]
- (ii) Suggest the structure of compound **D**. [1]

[Total: 21]

3 This question is about reactions involving some *d*-block elements and their ions.

- (a) Ammonia is a reagent commonly used to react with copper and silver ions. The metal ions can be precipitated from their solutions in some instances.

With the aid of relevant equations, explain the following observations as fully as you can.

- (i) When aqueous ammonia is added to a solution containing copper(II) ions, a pale blue precipitate is obtained. This precipitate dissolves to form a deep blue solution when excess ammonia is added. [3]
- (ii) When aqueous ammonia is added to AgCl , the precipitate dissolves. However, if aqueous ammonia is added to AgBr , the precipitate remains insoluble. [3]
- (b) Iron forms compounds mainly in the +2 and +3 oxidation states. Iron complexes are usually coloured.

- (i) Explain why iron exhibits variable oxidation states. [2]
- (ii) State the full electronic configuration of an iron(III) ion and draw fully-labelled diagrams of **all** the orbitals in the 3*d*-subshell of an iron(III) ion. [3]
- (iii) Using your diagrams in (b)(ii), briefly explain why splitting of the *d*-subshell occurs in an octahedral complex.

Hence, explain why iron(III) complexes are usually coloured. [3]

- (iv) In 2012, a group of scientists synthesised two iron(III) complexes with different colours. The iron(III) complex bonded to N,N-diethylethylenediamine ligands is red in colour and the one bonded to N-ethylethylenediamine ligands is blue in colour. The energy gap between the *d*-orbitals is inversely proportional to the wavelength of the light absorbed.

Given that red light has a longer wavelength than blue light, deduce whether the red complex or the blue complex has a larger energy gap between the *d*-orbitals. [2]

- (v) The arrangement of electrons in the *d*-orbitals depends on the spin states of complexes.

In a 'high spin' state, the electrons occupy all the *d*-orbitals singly, before starting to pair up in the lower energy *d*-orbitals.

In a 'low spin' state, the lower energy *d*-orbitals are filled first, by pairing up if necessary, before the higher energy *d*-orbitals are used.

It was found that the two iron(III) complexes in (b)(iv) have different 'spin' state for the Fe^{3+} ion.

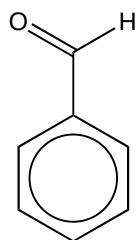
Use appropriate diagrams and your answer to (b)(iv) to show the electronic configuration of a Fe^{3+} ion in the red complex and the blue complex respectively. Your diagrams should indicate clearly the relative size of the energy gap for each complex. [2]

[Total: 18]

Section B

Answer **one** question from this section.

- 4 Benzaldehyde contains an aromatic benzene ring and has the following structure.



- (a) Benzaldehyde undergoes disproportionation reaction in the presence of a strong base such as potassium hydroxide to produce potassium benzoate and phenylmethanol. Potassium benzoate can be acidified to obtain benzoic acid as one of the final products.

The disproportionation reaction is as follows:

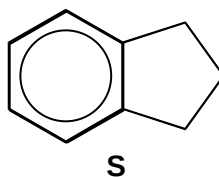


The rate of disappearance of potassium hydroxide was measured for several different concentrations of benzaldehyde and potassium hydroxide at a certain temperature.

experiment number	$[\text{C}_6\text{H}_5\text{CHO}]$ / mol dm^{-3}	$[\text{KOH}]$ / mol dm^{-3}	rate of disappearance of KOH / $\text{mol dm}^{-3} \text{ s}^{-1}$
1	0.300	0.050	5.7×10^{-5}
2	0.300	0.100	1.1×10^{-4}
3	0.600	0.200	9.2×10^{-4}

- (i) With reference to the above reaction, explain why it is a disproportionation reaction. [1]
- (ii) From the experimental data given in the table, determine the order of reaction with respect to $\text{C}_6\text{H}_5\text{CHO}$ and KOH . Explain your reasoning. [2]
- (iii) Hence write the rate equation for the disproportionation of benzaldehyde, and calculate the rate constant, giving its units. [2]
- (iv) Calculate the half-life of the reaction when the concentration of $\text{C}_6\text{H}_5\text{CHO}$ is increased to 1.60 mol dm^{-3} . [2]
- (v) With the aid of a sketch of the Boltzmann distribution, explain the effect of increasing temperature on the rate of reaction. [3]
- (b) The disproportionation reaction in (a) produces a low yield of phenylmethanol and benzoic acid.
- Two other separate reactions could be carried out to convert benzaldehyde to phenylmethanol and to benzoic acid.
- State the reagents and conditions needed to convert benzaldehyde to
- (i) phenylmethanol, and [1]
- (ii) benzoic acid. [1]

- (c) Suggest a chemical test that could be used to distinguish between benzaldehyde and phenylmethanol. You should state what you would observe for **each** compound. [2]
- (d) Aromatic hydrocarbon **R** and **S** are structural isomers, with molecular formula C_9H_{10} .
The structure of **S** is shown below.



R is found to exhibit stereoisomerism. When aqueous bromine was added to both **R** and **S** separately, only **R** decolourised orange aqueous bromine. When hot acidified potassium manganate(VII) solution was added to **R** and **S** separately, both decolourised purple potassium manganate(VII) to form compounds **T** and **U** respectively.

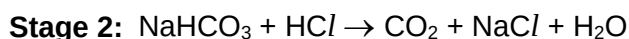
1.0 mol of compound **T** reacts with exactly 0.5 mol of sodium carbonate while 1.0 mol of compound **U** reacts with exactly 1.0 mol of sodium carbonate.

Deduce the structures of compounds **R**, **T** and **U**. Explain the chemistry of the reactions described. [6]

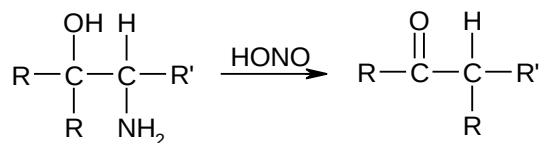
[Total: 20]

5 This question is about reactions involving acids and bases.

- (a) (i) Draw a dot-and-cross diagram to show the bonding in a carbonate anion, CO_3^{2-} .
You should distinguish carefully between electrons originating from the central atom and those from the outermost atoms.
Include all lone pairs in your diagram. [1]
- (ii) Use the VSEPR (valence shell electron pair repulsion) theory to predict the shape of the carbonate ion, giving a reason for your answer. [2]
- (b) The carbonate ion is a diacidic base. When sodium carbonate reacts with hydrochloric acid, the neutralisation reaction occurs in two stages.



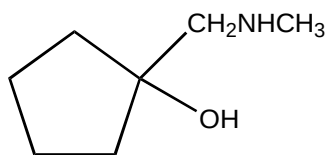
- (i) Explain why **Stage 1** is considered to be a Bronsted-Lowry acid-base reaction. [1]
- (ii) Explain, with the help of an equation, why the pH at the end of **stage 1** is slightly higher than 7. [2]
- (c) The major buffer system that is used to control the pH of blood is the carbonic acid, H_2CO_3 , and hydrogen carbonate, HCO_3^- , buffer system.
Explain, with the aid of relevant equations, how the pH of blood is maintained when a small amount of acid or alkali is added. [2]
- (d) (i) State **two** assumptions of the kinetic theory of gases. [2]
- (ii) When excess hydrochloric acid is added to a 20.0 cm^3 sample of aqueous sodium carbonate, 56.0 cm^3 of carbon dioxide gas was collected at 1.1 atm and 30°C .
Determine the concentration of the sample of aqueous sodium carbonate. [2]
- (e) A carbonyl compound, **H**, $\text{C}_6\text{H}_{10}\text{O}$, can be synthesised from 1-aminoalcohol, **G**, $\text{C}_6\text{H}_{13}\text{ON}$, in the presence of nitrous acid, HONO , via Tiffeneau-Demjanov Rearrangement. The simplified illustration of the rearrangement is illustrated below.



H produces a yellow precipitate with iodine in alkaline solution. Treatment of **H** with hot acidified solution of potassium manganate(VII) produces **J**, $\text{C}_5\text{H}_8\text{O}_3$, along with a gas that forms white precipitate in limewater. **H** was also observed to decolourise bromine in tetrachloromethane readily.

Explain the chemistry of the reactions described and deduce the structures of **G**, **H** and **J**. [5]

- (f) Starting from a suitable carbonyl compound of your choice, devise a 3-step synthesis of the compound below.



Suggest reagents and conditions for each step, and draw the structural formula of every intermediate compound. [3]

[Total: 20]

~ END OF PAPER ~